Lab 07: Run Deep Descriptive and Diagnostic Analysis in Excel

Topic 02 | Intermediate | Excel + Copilot | K3 | A2

Goal

Move from summary statistics to segmented drivers, anomalies and evidence-backed explanations.

Expected output

Diagnostic workbook with KPI summary, segment analysis, anomaly log and management findings.

Files

- Lumina-Living-Lab-07.xlsx - learner starter and control workbook
- completed/Lumina-Living-Lab-07-completed.xlsx - completed comparison copy
- LAB-07-Instructions.pdf - printable detailed guide
- Other Office files in this folder are specific to this lab

Primary prompt

Analyse the Campaign_Data table in four passes: descriptive, segmented, anomaly and diagnostic. Calculate volume, revenue, margin, conversion and return metrics; compare channel, region and product; flag unusual values using robust rules; then propose three likely drivers. Separate observed evidence from hypotheses and cite the supporting table or chart.

Procedure

1. Confirm the data table grain and period.

One row must represent one campaign-week-product combination.

2. Ask Copilot for a staged analysis plan.

Analyse the Campaign_Data table in four passes: descriptive, segmented, anomaly and diagnostic. Calculate volume, revenue, margin, conversion and return metrics; compare channel, region and product; flag unusual values using robust rules; then propose three likely drivers. Separate observed evidence from hypotheses and cite the supporting table or chart.

3. Build descriptive KPI formulas and reconcile totals.

Use SUMIFS, COUNTIFS, AVERAGEIFS, MEDIAN and STDEV.S where appropriate.

4. Create PivotTables for channel, region and product.

Show revenue, margin, conversion and return rate with consistent number formats.

5. Create trend and variance charts.

Use native Excel charts and label period and unit.

6. Investigate anomalies.

Record the row key, rule, observed value, possible cause and verification owner.

7. Ask Copilot to critique the findings.

> Challenge these findings. Identify alternative explanations, missing data and any claim that is stronger than the evidence.

8. Write the management findings.

Each finding must include evidence, caveat, action and owner.

Acceptance test

The workbook separates facts from hypotheses, reconciles totals, documents anomalies and presents three defensible findings with caveats.

Evidence

- Completed workbook with Prompt_Log and Evidence_Register
- App-specific artifact and a credential-safe screenshot
- Verification, exception and human-approval note

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